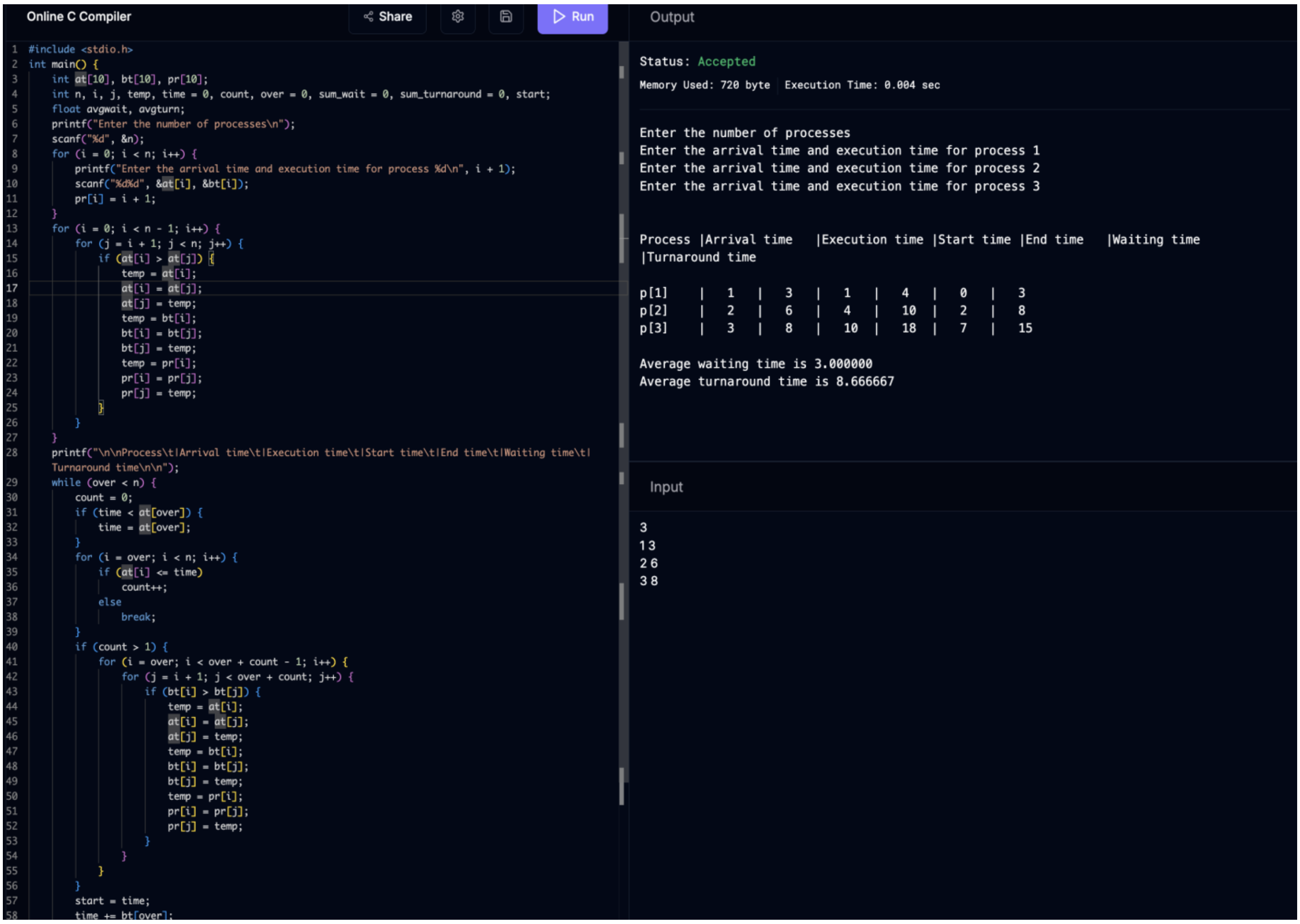
EXPERIMENT 7:

7. Construct a C program to implement a non-preemptive SJF algorithm.



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```c
#include <stdio.h>
int main() {
    int at[10], bt[10], pr[10];
    int n, i, j, temp, time = 0, count, over = 0, sum_wait = 0, sum_turnaround = 0, start;
    float avgwait, avgturn;
    printf("Enter the number of processes\n");
    scanf("%d", &n);
    for (i = 0; i < n; i++) {
        printf("Enter the arrival time and execution time for process %d\n", i + 1);
        scanf("%d%d", &at[i], &bt[i]);
        pr[i] = i + 1;
    }
    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {
            if (at[i] > at[j]) {
                temp = at[i];
                at[i] = at[j];
                at[j] = temp;
                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
                temp = pr[i];
                pr[i] = pr[j];
                pr[j] = temp;
            }
        }
    }
    printf("\n\nProcess\t|Arrival time\t|Execution time\t|Start time\t|End time\t|Waiting time\t|
    Turnaround time\n\n");
    while (over < n) {
        count = 0;
        if (time < at[over]) {
            time = at[over];
        }
        for (i = over; i < n; i++) {
            if (at[i] <= time)
                count++;
            else
                break;
        }
        if (count > 1) {
            for (i = over; i < over + count - 1; i++) {
                for (j = i + 1; j < over + count; j++) {
                    if (bt[i] > bt[j]) {
                        temp = at[i];
                        at[i] = at[j];
                        at[j] = temp;
                        temp = bt[i];
                        bt[i] = bt[j];
                        bt[j] = temp;
                        temp = pr[i];
                        pr[i] = pr[j];
                        pr[j] = temp;
                    }
                }
            }
        }
        start = time;
        time += bt[over];
```

Output

```
Status: Accepted
Memory Used: 720 byte   Execution Time: 0.004 sec

Enter the number of processes
Enter the arrival time and execution time for process 1
Enter the arrival time and execution time for process 2
Enter the arrival time and execution time for process 3


Process |Arrival time   |Execution time |Start time |End time   |Waiting time
|Turnaround time

p[1]    |   1   |   3   |   1   |   4   |   0   |   3
p[2]    |   2   |   6   |   4   |   10  |   2   |   8
p[3]    |   3   |   8   |   10  |   18  |   7   |   15

Average waiting time is 3.000000
Average turnaround time is 8.666667
```

Input

```
3
1 3
2 6
3 8
```